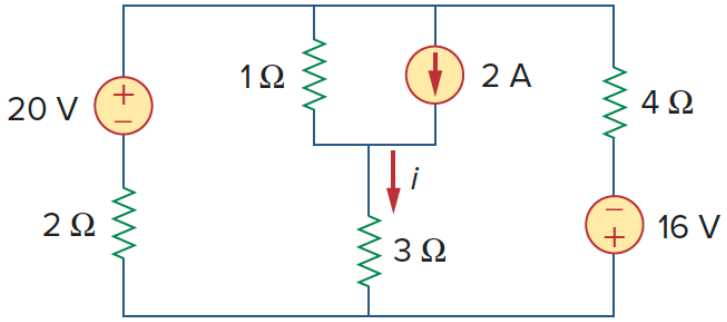
Lab 04 Superposition and substitution Theorem (submit before Feb. 23)

All tasks can be conducted on CircuitLab

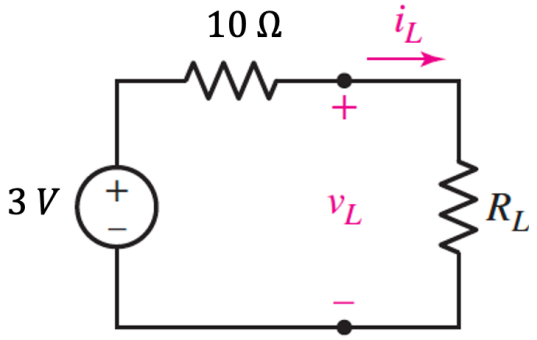
Task 1: This task aims to verify the **Superposition** Theorem.
Build the circuit shown in below in CircuitLab.
Activate one source at a time (turn off the others properly), record all data in the table, and paste your circuit schematic below.



	Current i
All sources are acting	
20 V voltage source is acting alone	
16 V voltage source is acting alone	
2 A current source is acting alone	

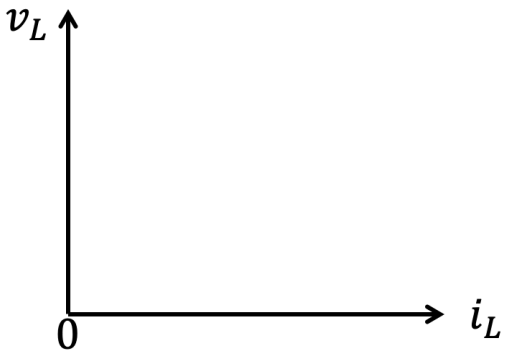
Paste your circuits down:

Task 2: Build the circuit shown below in CircuitLab. Vary the load resistance from 0 Ω to 50 Ω, and record the corresponding voltages and currents.



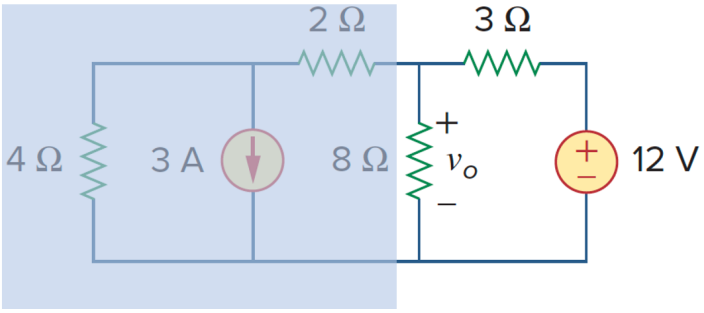
R_L	0 Ω	5 Ω	20 Ω	50 Ω
v_L				
i_L				

Plot v_L versus i_L in the following coordinate axes using your recorded data. Then derive the mathematical expression that describes the v–i relationship.



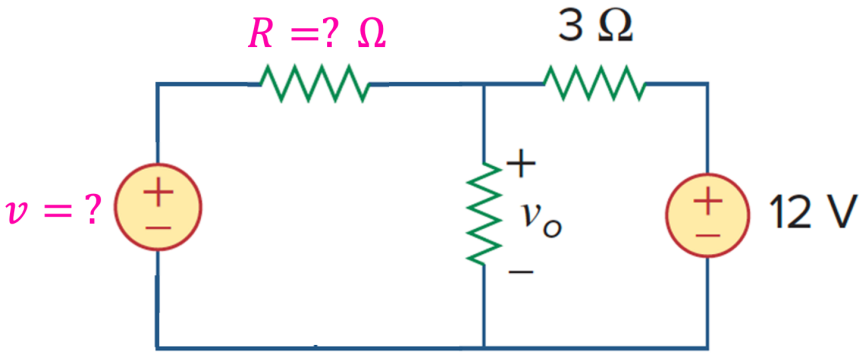
$v_L =$ _____

Task 3: Source Transformation.
Build the following circuit in CircuitLab, and record the voltage across 8Ω resistor v_0.



$v_0 =$ _____

Use source transformation to transform the left part circuit (covered in blue square) into the following form. Build the resulting circuit in CircuitLab, and record the voltage v_0 again.



$R =$ _____

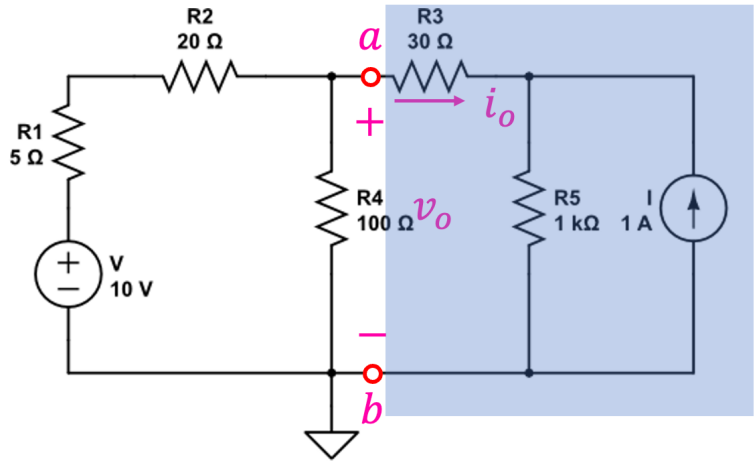
$v =$ _____

$v_0 =$ _____

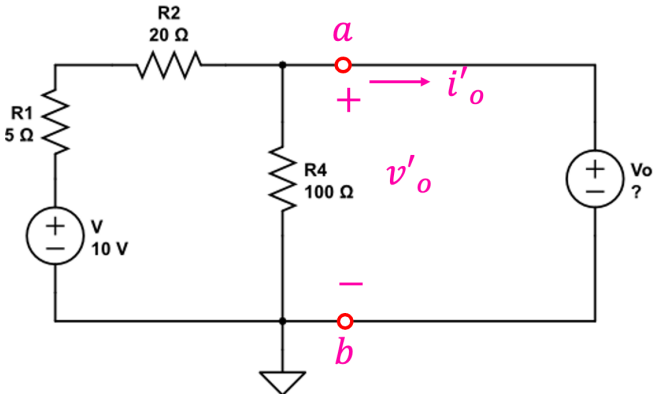
Compare the two values of v_0, are they equal? What conclusion can you draw?

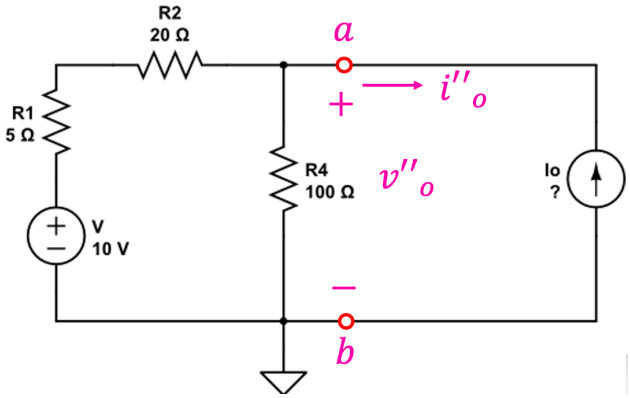
Answer:

Task 4: Substitution Theorem.



Build the circuit shown above in CircuitLab. Record the voltage between node a and b, i.e. v_o, and the current flowing through R3,, $i.e.\ i_o$.
Substitute the right part circuit (covered in blue square) by a voltage or current source in CircuitLab. Record all data in the following table.





	value
v_o	
v'_o	
v''_o	

Are these value equivalent? What conclusion can you draw?

Answer: